

RITIK SAH

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PROFESSIONAL SUMMARY

Computing Systems undergraduate and software engineer building reliable digital systems across cloud, full-stack, AI-assisted and security-focused work. Public evidence includes an OWASP open-source maintainer role, an externally published AI-security integration, and tested network-security and threat-intelligence platforms. Brings practical Python, Azure, API, data and delivery experience, with a focus on clear system boundaries, automated verification and software that remains understandable to users and contributors.

PROJECTS

OpenShield | API & AI Maintainer | Infrastructure Lead | OWASP Open Source

- Cloud teams need repeatable evidence of Azure posture risks; OpenShield scans environments for misconfigurations and quantum-unsafe cryptography.
- Maintains API and AI areas, built GitHub Actions quality gates, connected the Python and Flask API to the React dashboard, and supported PostgreSQL-backed delivery.
- Public evidence includes the OWASP maintainer record and an open repository where contribution, review and delivery work can be inspected.

IoT Sentinel | Privacy-first Raspberry Pi Network-Security Appliance

- Home-network users need understandable, locally controlled security visibility; IoT Sentinel discovers devices, monitors traffic and guides investigation.
- Implemented Zeek analysis, River online learning, guarded firewall enforcement and an Ollama-first privacy mode; validated 1,263 tests across 54 files at audit.
- Designed a five-step investigation pipeline and configurable six-tier AI fallback so degraded providers do not remove core local monitoring.

DUSK | Jointly Built AI Security Evaluation Framework

- AI-agent activity needs explainable behavioural evaluation; DUSK maps security-relevant findings to MITRE ATT&CK and MITRE ATLAS.
- Co-built the framework and integrated it into Superlinked SIE as agent-action-monitor, including tests, sample data and a Docker Compose demonstration.
- The repository benchmark reports 1.0 precision, 1.0 recall and 0.0 false-positive rate for that scoped benchmark, not universal production performance.

BreachLens | Full-stack Cyber Threat Intelligence Platform

- Raw breach feeds are difficult to navigate and risky to query; BreachLens turns them into searchable intelligence with privacy-preserving checks.
- Built Angular 17 and Flask services with MongoDB, 65 REST endpoints, automated OSINT synchronisation, maps and dashboards, protected by 889 tests at audit.
- Designed a 1.7 MB in-memory Bloom Filter for millions of exposed-email signatures and added k-anonymity checks to reduce unnecessary disclosure.

ThreatVault | Cloud-native Digital-Evidence Vault | Azure Coursework

- Digital evidence needs controlled storage, moderation and an inspectable audit trail; ThreatVault separates binary evidence, threat indicators and user activity across Blob Storage, Azure SQL and Cosmos DB.
- Orchestrated CRUD operations through 19 Azure Logic App workflows, protected secrets with Azure Key Vault, and added Azure AI Content Safety plus audit-volume anomaly checks.

SKILLS

Programming and APIs: Python, Java, SQL, Flask, REST APIs, Angular 17

Cloud and delivery: Microsoft Azure, Docker, Linux, GitHub Actions, Key Vault, Logic Apps

Data and systems: PostgreSQL, MongoDB, Raspberry Pi, Zeek, River ML, Ollama

EXPERIENCE

Database Administrator | Arunima Secondary School | Part-time, Hybrid | Kathmandu | Dec 2020 to Jul 2022

- Managed SQL and Oracle records, supporting accurate organisation, maintenance and retrieval of administrative data.
- Analysed data trends and prepared actionable findings to support operational decisions.
- Used data-entry and spreadsheet tools to strengthen record accuracy, consistency and technical precision.

EDUCATION

BSc (Hons) Computing Systems with Diploma in Professional Practice | Ulster University, London

Expected First-Class Honours; Dean's Award recipient in Years 1 and 2. Winner, QAHE TechFest 2026 Ulster Project Showcase and Poster Exhibition. Relevant study includes database systems, systems security, networking and cloud-native development.